

RITIK GUPTA

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Summary

Full-stack and applied AI engineer building end-to-end web applications across React/Vue frontends, Python/Node.js backends, databases, APIs, and cloud infrastructure, with a focus on responsive and user-focused digital products.

Education

Indian Institute of Technology Madras

Sep 2023 – Nov 2025

B.S. (Diploma in Programming)

Chennai, India

Chhatrapati Sahuji Maharaj University

Aug 2019 – Sep 2022

B.Sc. in Mathematics, Physics & Economics

Kanpur, India

Projects

KrishiRakshak — AI Crop & Livestock Doctor | Flask, Gemini 2.5 Flash, AWS S3, DynamoDB, EC2, PWA 2026

- Built a bilingual (Hindi/Marathi) multimodal disease-detection app for the AWS AI for Bharat Hackathon, targeting low-connectivity rural users.
- Engineered a vision-language pipeline using Gemini 2.5 Flash with two-stage prompt chaining (image validation and structured JSON output); integrated OpenWeather API for weather-based risk alerts; architected AWS backend (S3 + DynamoDB) with client-side image compression for offline-capable PWA delivery.
- Delivered a live, EC2-deployed prototype with a Chart.js analytics dashboard tracking diagnosis history and severity trends; accepted as a qualified hackathon submission. [GitHub](#)

Quiz Master — AI-Powered Adaptive Quiz Platform | Flask, PostgreSQL, Supabase, Gemini API, Tailwind, Render 2026

- Generic quiz tools lack adaptive AI generation, cost reliability, and concurrent multi-user isolation.
- Developed a production-grade, multi-role quiz platform with zero-cost AI generation and robust session handling.
- Designed an 8-model free-tier fallback chain across Gemini variants to ensure 99%+ AI uptime; implemented SHA-256 SQL response caching to eliminate redundant LLM calls; enforced anti-cache-poisoning headers and session scoping for multi-tenant data isolation; built XP-based leaderboards and real-time Chart.js analytics.
- Platform live on Render with Supabase-hosted PostgreSQL supporting multi-user concurrent sessions; caching measurably reduced repeat-query latency. [Demo](#) | [GitHub](#)

Smart Parking Management System | Flask, Vue.js, Redis, Celery, SQLite, Bootstrap, Render 2025

- Built a full-stack role-based application featuring responsive design, reservation workflows, dynamic pricing, Redis caching, Celery background jobs, and admin/user dashboards; designed and verified REST APIs via Postman. [Demo](#) | [GitHub](#)

Technical Skills

Languages & Frameworks: Python, JavaScript (ES6+), TypeScript, SQL, Flask, React.js, Node.js, Tailwind CSS

AI & LLM: Google Gemini API, Prompt Engineering, Prompt Chaining, Multi-Agent Systems, RAG, Gemini ADK

Cloud & Databases: AWS (EC2, S3, DynamoDB), Google Cloud, PostgreSQL, Supabase, Redis

DevOps & Tools: Docker, GitHub Actions, CI/CD, Celery, Postman, Git, Vercel, Render, Linux/WSL

Experience & Certifications

Full-Stack Web Development Intern — Jyesta Corporate Entity Sep – Nov 2025

- Strengthened full-stack development fundamentals through hands-on practice with React.js, Node.js, JavaScript (ES6+), SQL, and Git/GitHub during a structured virtual internship.
- Built and tested REST APIs, CRUD workflows, responsive UI components, and database operations while following modern software engineering practices and version control workflows.
- Improved understanding of frontend-backend integration, debugging, authentication concepts, database design, and collaborative development using Git-based workflows.
- Google Cloud Career Launchpad: Generative AI Leader — Google Cloud Jan–Apr 2026
Foundation models, RAG, AI agents, responsible AI on GCP. [View](#)
- AWS Academy Graduate: Cloud Foundations — AWS May 2026
Core AWS services, infrastructure management, and cloud security basics. [View](#)